6.5.5 Downstream Processing

The various processes used for the actual recovery of useful products from fermentation or any other industrial processes are called downstream processing. The cost of downstream processing (DSP) is often more than 50% of the manufacturing cost, and there is product loss at each step of DSP. Therefore, the DSP should be efficient, involve as few steps as possible and be cost-effective. Methods involved in the downstream processing are outlined in the flowchart (6.2). Table 6.4 shows Difference between upstream and downstream processing.

Table 6.4: Difference between upstream (usp) and downstream (dsp) processing

USP	DSP
USP overall makes the procurement and maintenance of inoculum	DSP depends upon selection of cost–effective media
USP involves in strain improvement to enhance and yield	DSP concentrates on media optimization for maximum productivity yield and profit
It is a continuous development of selected strains to increase the economic yield	For DSP, fermentation conditions are optimized for the growth of micro organism or the production of a desired product

6.6 Penicillin Production

Penicillin is a broad spectrum antibiotic. Penicillin is first obtained from the mould, *Penicillium notatum* (Figure 6.9).

Penicillium chrysogenum is a high yielding strain, used for the commercial production of penicillin. This strain is highly unstable,

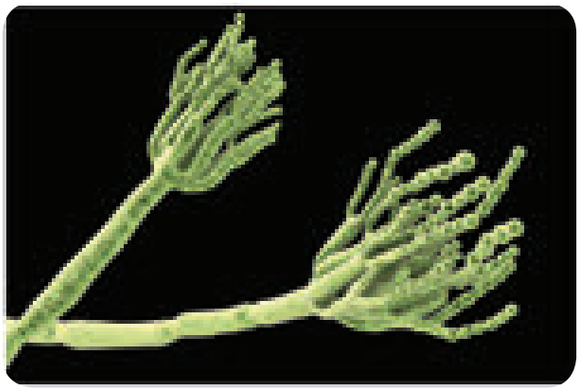


Figure 6.9: Structure of *Penicillium notatum*

so the spore suspensions are maintained in a dormant state to prevent contamination. Most penicillin form filamentous broth and hence is difficult to mix and it hinders oxygen transfer due to their high viscosity. This is avoided by using bubble columns air lift reactors which agitates the medium providing even oxygen distribution.

Infobits

In later (1939) using (Flemings' work) Howard Florey and Ernst Chain managed to purify penicillin in a powdered form. In 1941, they successfully treated a human. In 1943, they produced penicillin on a large scale. This helped immensely to treat causalities during the 2nd World War ww1 that had bacterial inflations due to their wounds.

Penicillin has a basic structure 6-amino penicillanic acid 6-(APA). It consists of a thiazolidine ring with a condensed β-lactum ring. It carries a variable side chain in position 6. Natural penicillins are produced in a fermentation process without adding any side chain precursors. If a side chain precursor is added to the broth, desired penicillin is produced and it is called bio-synthetic penicillin.

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